

Software Requirements Specification

for

E-Commerce Web Application

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Prepared based on	System Entity-Relationship Diagram (ERD.png)

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements of the **E-Commerce Web Application**. The purpose of this document is to provide a clear and detailed description of the system's behavior, data structures, and constraints, derived from the system's Entity-Relationship Diagram (ERD), to guide the design, development, and testing of the application.

1.2 Scope

The E-Commerce Web Application enables customers to browse product categories and products, manage a personal shopping cart, place orders, and rate/review purchased products. It also supports administrative management of categories and products. The system covers the following core modules: User Management, Category Management, Product Management, Shopping Cart, Order Management, and Product Rating & Review.

1.3 Definitions, Acronyms and Abbreviations

Term	Description
SRS	Software Requirements Specification
ERD	Entity-Relationship Diagram
PK	Primary Key
FK	Foreign Key
UK	Unique Key / Unique Constraint
CRUD	Create, Read, Update, Delete
User	A registered customer of the system
Admin	A privileged user who manages catalog data

1.4 References

1.5 Overview

The remainder of this document is organized as follows: Section 2 gives an overall description of the product and its users; Section 3 lists detailed functional requirements grouped by module; Section 4 describes non-functional requirements; Section 5 documents the data model derived from the ERD; Section 6 summarizes key use cases; and Appendix A presents the full ERD.

2. Overall Description

2.1 Product Perspective

The E-Commerce Web Application is a standalone, self-contained web-based system consisting of a customer-facing storefront and back-end data management for products, orders, and users. It follows a relational data model, as depicted in the system ERD, comprising eight core entities: **User**, **Category**, **Product**, **Cart**, **CartItem**, **Order**, **OrderItem**, and **Rating**.

2.2 Product Functions

- User registration, authentication, and account management
- Browsing of product categories and products
- Management of a personal shopping cart (add, update, remove items)
- Placing and tracking orders, including shipping details and payment status
- Submitting one rating and comment per product per user
- Administrative management of categories and products

2.3 User Classes and Characteristics

User Class	Description
Customer (User)	A registered user who browses products, manages a cart, places orders, and rates products.
Administrator	A user with elevated privileges responsible for managing categories, products, and order fulfillment.
Guest	An unauthenticated visitor who can browse categories and products but must register/log in to purchase.

2.4 Operating Environment

The system operates as a web application accessible via standard web browsers, backed by a relational database that implements the entities and relationships defined in Section 5.

2.5 Design and Implementation Constraints

- Each **User** has at most one **Cart** (one-to-one relationship).

- Each **Rating** is constrained by a unique combination of `product_id` and `user_id` (a user may rate a given product only once).
- **Category** and **Product** slugs must be unique across the system.
- All monetary values (e.g., `Product.price` , `OrderItem.price`) must be stored using a decimal data type to avoid rounding errors.

2.6 Assumptions and Dependencies

It is assumed that a single default currency is used throughout the system, that products belong to exactly one category, and that payment processing/transaction verification is handled by an external payment gateway referenced via the `Order.transaction_id` field.

3. Functional Requirements

3.1 User Management

ID	Requirement
FR-1.1	The system shall allow a new user to register with a unique username, email, and password.
FR-1.2	The system shall allow a registered user to log in using their username/email and password.
FR-1.3	The system shall securely store user passwords (e.g., hashed, never in plain text).
FR-1.4	The system shall automatically associate exactly one shopping cart with each registered user.

3.2 Category Management

ID	Requirement
FR-2.1	The system shall allow an administrator to create, update, and delete product categories.
FR-2.2	Each category shall have a unique name, a unique slug, and an optional description.
FR-2.3	The system shall allow users to browse products filtered by category.

3.3 Product Management

ID	Requirement
FR-3.1	The system shall allow an administrator to create, update, and delete products.
FR-3.2	Each product shall belong to exactly one category and include name, unique slug, description, price, stock quantity, availability flag, and image.
FR-3.3	The system shall automatically record the creation and last-updated timestamps of each product.
FR-3.4	The system shall prevent a product from being purchased when its stock is zero or its availability flag is false.

FR-3.5	The system shall allow users to view detailed information and ratings for a given product.
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3.4 Shopping Cart

ID	Requirement
FR-4.1	The system shall provide each logged-in user with exactly one persistent cart.
FR-4.2	The system shall allow a user to add a product to their cart as a cart item with a specified quantity.
FR-4.3	The system shall allow a user to update the quantity of, or remove, a cart item.
FR-4.4	The system shall record and update the cart's <code>created_at</code> and <code>updated_at</code> timestamps.

3.5 Order Management

ID	Requirement
FR-5.1	The system shall allow a user to place an order, converting the current cart items into order items.
FR-5.2	Each order shall capture recipient details: first name, last name, email, address, postal code, and city.
FR-5.3	Each order shall have a status (e.g., pending, processing, shipped, delivered, cancelled) and an optional note.
FR-5.4	The system shall track whether an order has been paid and store an associated payment transaction ID.
FR-5.5	Each order item shall record the product, quantity, and the price of the product at the time of purchase.
FR-5.6	The system shall allow a user to view their order history and the current status of each order.

3.6 Product Rating & Review

ID	Requirement
FR-6.1	The system shall allow a user to submit a numeric rating and an optional comment for a product.
FR-6.2	The system shall restrict each user to at most one rating per product (enforced by a unique product-user constraint).
FR-6.3	The system shall record the timestamp when a rating is created.
FR-6.4	The system shall display the ratings and comments associated with a product on its detail page.

4. Non-Functional Requirements

4.1 Performance

The system shall load product listing and product detail pages within 2 seconds under normal load conditions.

4.2 Security

- User passwords shall be stored using a strong one-way hashing algorithm.
- All sensitive operations (checkout, account management) shall require an authenticated session.
- Payment transaction identifiers shall not be exposed to unauthorized users.

4.3 Usability

The user interface shall be intuitive and require no special training for a typical online shopper to browse products, manage a cart, and place an order.

4.4 Reliability & Availability

The system shall ensure that order and payment data is never lost once an order has been confirmed, and shall target at least 99% uptime.

4.5 Scalability

The data model shall support growth in the number of categories, products, users, and orders without requiring structural changes to the schema.

4.6 Maintainability

The system's data access layer shall reflect the entity relationships defined in Section 5, allowing future features to be added with minimal impact on existing modules.

5. Data Requirements (Entity-Relationship Model)

This section documents the core entities and their attributes as derived from the system ERD (see Appendix A).

5.1 User

Field	Type	Key
id	int	PK
username	string	
email	string	
password	string	

5.2 Category

Field	Type	Key
id	int	PK
name	string	

slug	string	UK
description	string	

5.3 Product

Field	Type	Key
id	int	PK
name	string	
slug	string	UK
category_id	int	FK → Category.id
description	string	
price	decimal	
stock	int	
available	boolean	
created	datetime	
updated	datetime	
image	string	

5.4 Cart

Field	Type	Key
id	int	PK
user_id	int	FK → User.id (One-to-One)
created_at	datetime	
updated_at	datetime	

5.5 CartItem

Field	Type	Key
id	int	PK
cart_id	int	FK → Cart.id
product_id	int	FK → Product.id
quantity	int	

5.6 Order

Field	Type	Key
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id	int	PK
user_id	int	FK → User.id
first_name	string	
last_name	string	
email	string	
address	string	
postal_code	string	
city	string	
created	datetime	
updated	datetime	
status	string	
note	string	
paid	boolean	
transaction_id	string	

5.7 OrderItem

Field	Type	Key
id	int	PK
order_id	int	FK → Order.id
product_id	int	FK → Product.id
price	decimal	
quantity	int	

5.8 Rating

Field	Type	Key
id	int	PK
product_id	int	FK → Product.id
user_id	int	FK → User.id
rating	int	
comment	string	
created	datetime	
unique_product_user	constraint	UK (product_id, user_id)

5.9 Entity Relationships Summary

Relationship	Cardinality
Category → Product	One category has many products
User → Cart	One user has one cart (1:1)
User → Order	One user places many orders
Cart → CartItem	One cart contains many cart items
Product → CartItem	One product is included in many cart items
Order → OrderItem	One order contains many order items
Product → OrderItem	One product is included in many order items
Product → Rating	One product has many ratings
User → Rating	One user gives many ratings (at most one per product)

6. Use Case Summaries

UC-1: Browse Products by Category

Actor: Guest / Customer. **Description:** The user selects a category and views the list of available products belonging to it. **Precondition:** At least one category and product exist. **Postcondition:** A filtered product list is displayed.

UC-2: Add Product to Cart

Actor: Customer. **Description:** The user selects a product and quantity, and adds it to their cart. **Precondition:** User is logged in; product is available and in stock. **Postcondition:** A new CartItem is created or an existing one's quantity is updated.

UC-3: Place an Order (Checkout)

Actor: Customer. **Description:** The user reviews cart items, provides shipping details, and confirms the order. **Precondition:** Cart contains at least one item. **Postcondition:** An Order and its corresponding OrderItems are created; the cart is cleared.

UC-4: Rate a Product

Actor: Customer. **Description:** The user submits a rating (and optional comment) for a product they have purchased. **Precondition:** User has not already rated this product. **Postcondition:** A new Rating record is created and displayed on the product page.

UC-5: Manage Products (Admin)

Actor: Administrator. **Description:** The administrator creates, edits, or removes a product, including its category assignment, price, and stock. **Postcondition:** The Product catalog is updated.

Appendix A: Entity-Relationship Diagram

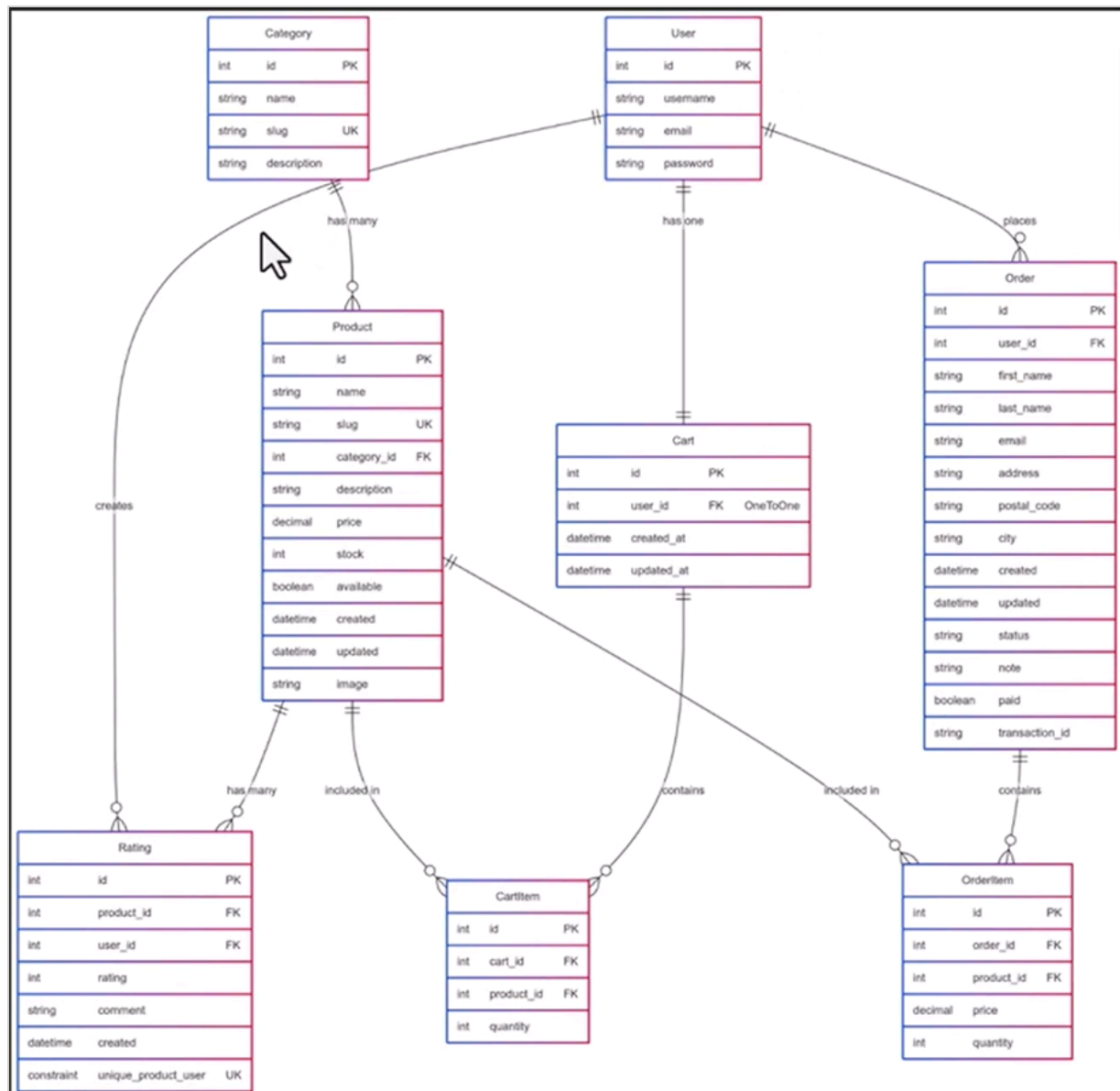


Figure A.1 — Entity-Relationship Diagram of the E-Commerce Web Application